

```
# Install Hugging Face Transformers and required dependencies
!pip install transformers[sentencepiece] datasets evaluate accelerate
```

```
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (24.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (6.0.1)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (2024.9.0)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.20.1)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.4.5)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (4.67.1)
Requirement already satisfied: sentencepiece!=0.1.92,>=0.1.91 in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (0.2.0)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from transformers[sentencepiece]) (5.29.6)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0]) (2025.3.0)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.10.0+cpu)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0]) (3.10.10)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (1.2.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->huggingface-hub) (0.27.0)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (4.12.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.3.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2025.1.1)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.6)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0]) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0]) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0]) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0]) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0]) (6.3.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0]) (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0]) (1.18.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.7.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.7)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.14.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=2.0.0) (3.1.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=2.0.0->accelerate) (3.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (2.18.1)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->transformers) (0.1.2)
Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
```

84.1/84.1 kB 1.9 MB/s eta 0:00:00

Installing collected packages: evaluate
Successfully installed evaluate-0.4.6

```
# Import necessary libraries for summarization tasks
from transformers import pipeline
import torch
import pandas as pd
```

```
print("Transformers library imported successfully!")
print(f"PyTorch version: {torch.__version__}")
```

Transformers library imported successfully!
PyTorch version: 2.10.0+cpu

```
# Initialize a summarization pipeline with a pretrained model
# facebook/bart-large-cnn is a popular choice for abstractive summarization of longer texts.
# Due to an environment issue where 'summarization' task is not recognized by pipeline, we will load the model and tokenizer di
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")

# We can still create a pipeline-like callable for convenience if needed, or use tokenizer/model directly
# summarizer = pipeline("summarization", model=model, tokenizer=tokenizer) # This would still fail due to the task name issue

# For now, we'll confirm direct loading
print("Tokenizer and Model for facebook/bart-large-cnn loaded successfully!")
```

vocab.json: 899k/? [00:00<00:00, 32.0MB/s]

merges.txt: 456k/? [00:00<00:00, 23.4MB/s]

tokenizer.json: 1.36M/? [00:00<00:00, 38.5MB/s]

model.safetensors: 100% 1.63G/1.63G [00:28<00:00, 60.3MB/s]

Please make sure the generation config includes `forced\_bos\_token\_id=0`.

Loading weights: 100% 511/511 [00:01<00:00, 552.57it/s, Materializing param=model.encoder.layers.11.self_attn_layer_norm.weight]

generation_config.json: 100% 363/363 [00:00<00:00, 14.0kB/s]

Tokenizer and Model for facebook/bart-large-cnn loaded successfully!

```
# Prepare diverse input texts for summarization
# We'll create a pandas DataFrame to store our sample texts along with their categories.

data = [
    {
        "category": "News Article",
        "text": "Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Ge
    },
    {
        "category": "Scientific Abstract",
        "text": "This paper presents a novel deep learning architecture for image segmentation using a combination of convoluti
    },
    {
        "category": "Product Review",
        "text": "I recently purchased the XYZ wireless earbuds, and I must say, I'm quite disappointed. The sound quality is me
    },
    {
        "category": "Conversational Text",
        "text": "Hey, did you catch the game last night? It was insane! The Lakers almost blew it in the fourth quarter, but Le
    },
    {
        "category": "Legal Document Excerpt",
        "text": "WHEREAS, the parties hereto desire to enter into this Agreement to define their respective rights and obligati
    }
]

df_texts = pd.DataFrame(data)
display(df_texts.head())

print(f"Prepared {len(df_texts)} texts from various domains.")
```

	category	text
0	News Article	Recent studies have shown a significant increa...
1	Scientific Abstract	This paper presents a novel deep learning arch...
2	Product Review	I recently purchased the XYZ wireless earbuds,...
3	Conversational Text	Hey, did you catch the game last night? It was...
4	Legal Document Excerpt	WHEREAS, the parties hereto desire to enter in...

Prepared 5 texts from various domains.

```
# Function to generate a summary for a given text, with customizable parameters
def generate_summary(text, max_length, min_length, do_sample=False, num_beams=4):
    # Tokenize the input text
    inputs = tokenizer(text, max_length=1024, truncation=True, return_tensors="pt")

    # Generate summary tokens
    summary_ids = model.generate(
        inputs["input_ids"],
```

```

        num_beams=num_beams,
        max_length=max_length,
        min_length=min_length,
        do_sample=do_sample
    )

    # Decode the generated tokens back to text
    summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
    return summary

# We will use the first text (News Article) for demonstrating different summary lengths
news_article_text = df_texts.loc[0, 'text']

print("Original News Article:")
print(news_article_text)
print("\n---")

# Generate a Short Summary
short_summary = generate_summary(news_article_text, max_length=50, min_length=10)
print("Short Summary (max_length=50, min_length=10):")
print(short_summary)
print("\n---")

# Generate a Medium Summary
medium_summary = generate_summary(news_article_text, max_length=100, min_length=30)
print("Medium Summary (max_length=100, min_length=30):")
print(medium_summary)
print("\n---")

# Generate a Long Summary
long_summary = generate_summary(news_article_text, max_length=150, min_length=50)
print("Long Summary (max_length=150, min_length=50):")
print(long_summary)
print("\n---")

# Generate a summary with do_sample=True (more creative/less deterministic)
sampled_summary = generate_summary(news_article_text, max_length=100, min_length=30, do_sample=True, num_beams=1) # num_beams=1
print("Sampled Summary (max_length=100, min_length=30, do_sample=True, num_beams=1):")
print(sampled_summary)
print("\n---")

```

```

Original News Article:
Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Germany and Sweden a

---
Short Summary (max_length=50, min_length=10):
Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Germany and Sweden a

---
Medium Summary (max_length=100, min_length=30):
Recent studies have shown a significant increase in renewable energy adoption across Europe. Experts predict that by 2030, a sub

---
Long Summary (max_length=150, min_length=50):
Recent studies have shown a significant increase in renewable energy adoption across Europe. Countries like Germany and Sweden a

---
Sampled Summary (max_length=100, min_length=30, do_sample=True, num_beams=1):
Germany and Sweden are leading Europe's adoption of cleaner energy. Cost of renewables has decreased, so cost is decreasing. Exp

---

```

```

summarized_texts = []

for index, row in df_texts.iterrows():
    category = row['category']
    original_text = row['text']

    # Generate a medium-length summary for consistency in comparison
    summary = generate_summary(original_text, max_length=100, min_length=30)

    summarized_texts.append({
        'category': category,
        'original_text': original_text,
        'generated_summary': summary
    })

```

```
df_summaries = pd.DataFrame(summarized_texts)
display(df_summaries)

print("Summaries generated for all diverse text types.")
```

	category	original_text	generated_summary
0	News Article	Recent studies have shown a significant increa...	Recent studies have shown a significant increa...
1	Scientific Abstract	This paper presents a novel deep learning arch...	'SegAttentionNet' achieves state-of-the-art pe...
2	Product Review	I recently purchased the XYZ wireless earbuds,...	The sound quality is mediocre at best, lacking...
3	Conversational Text	Hey, did you catch the game last night? It was...	The Lakers almost blew it in the fourth quarte...
4	Legal Document Excerpt	WHEREAS, the parties hereto desire to enter in...	The parties hereto desire to enter into this A...

Summaries generated for all diverse text types.

```
long_text = """The industrial revolution, which began in Great Britain in the late 18th century, marked a profound shift in hum

print("Original Long Text (first 200 characters):")
print(long_text[:200] + "...")
print(f"Total length of original text: {len(long_text.split())} words")

# Summarize the long text with a generous max_length to capture more information
long_text_summary = generate_summary(long_text, max_length=200, min_length=50)

print("\n--- Generated Summary ---")
print(long_text_summary)
print(f"Total length of summary: {len(long_text_summary.split())} words")
```

Original Long Text (first 200 characters):
The industrial revolution, which began in Great Britain in the late 18th century, marked a profound shift in human history, fund
Total length of original text: 314 words

--- Generated Summary ---
The industrial revolution began in Great Britain in the late 18th century. It was characterized by the transition from agrarian
Total length of summary: 48 words